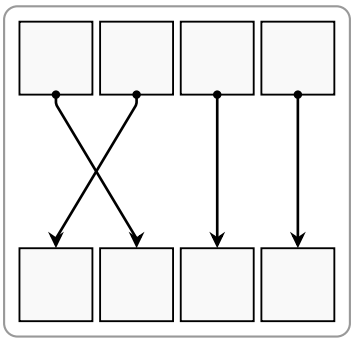
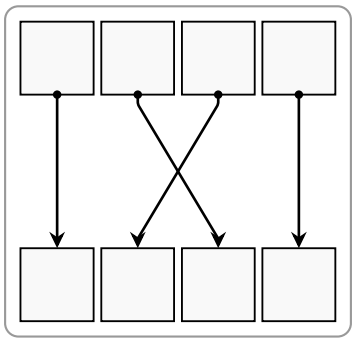
One turn: move every cube down its own arrow, then slide the whole bottom row straight up into the top row. When you draw a machine, give every top slot one arrow going out and every bottom slot one arrow coming in. A cube is home when it is back under its own picture.

Problem 6: One turn of machine A followed by one turn of machine B moves the cubes the same way as one turn of a single machine, called A then B. For each pair of machines A and B, draw A then B and B then A.

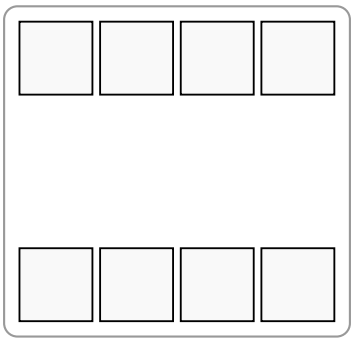
A



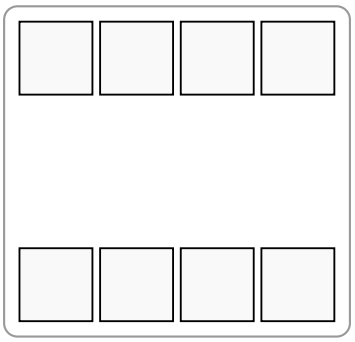
B



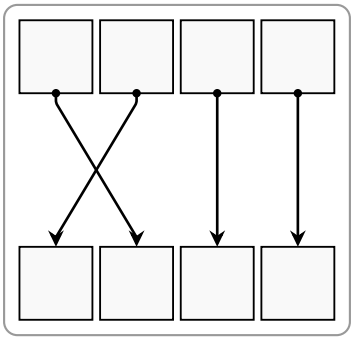
A then B



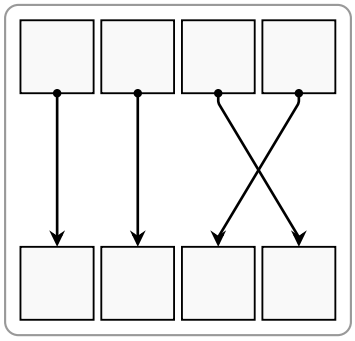
B then A



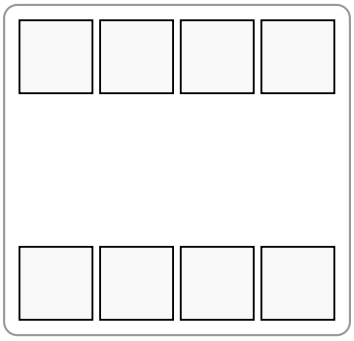
A



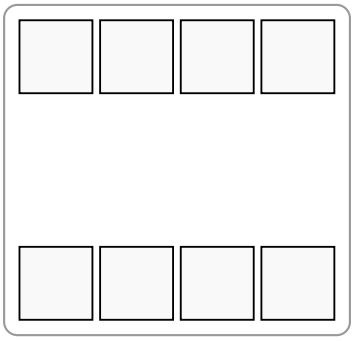
B



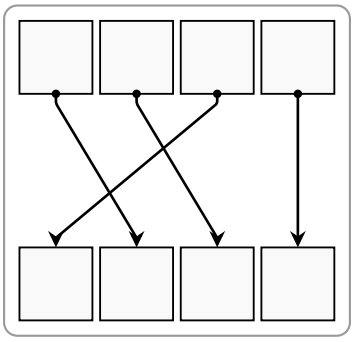
A then B



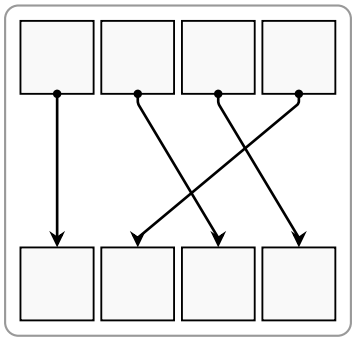
B then A



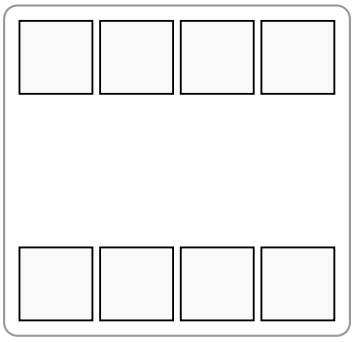
A



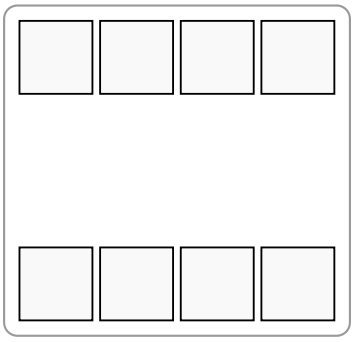
B



A then B

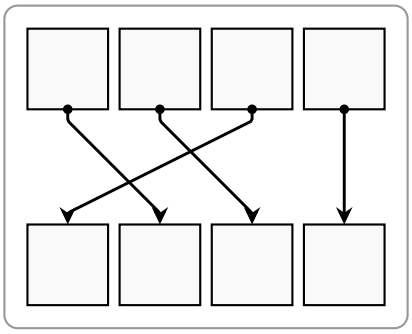


B then A

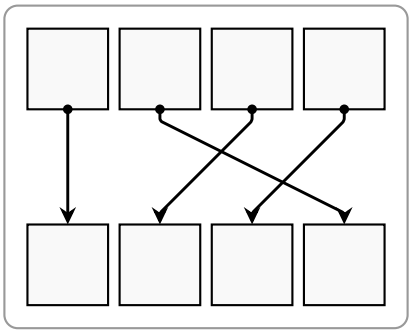


Problem 7: A machine undoes A if A then that machine puts every cube home. For each pair of machines A and B below, draw A then B, and draw the machines that undo A, B and A then B. Explain how to get the machine that undoes A then B from the machines that undo A and B, for any two machines A and B.

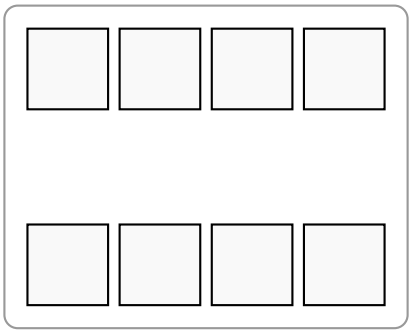
A



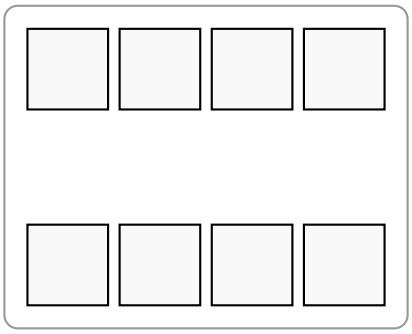
B



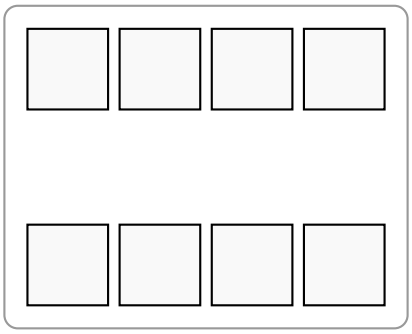
A then B



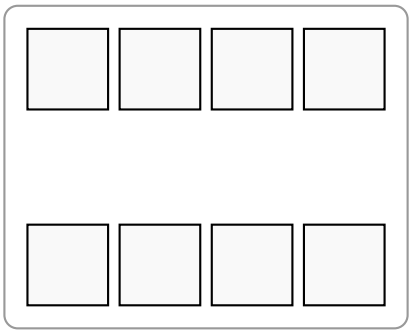
undoes A



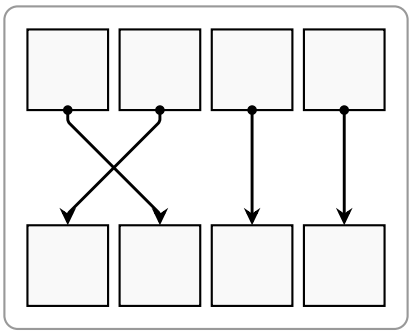
undoes B



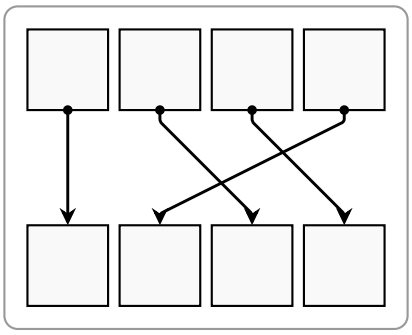
undoes (A then B)



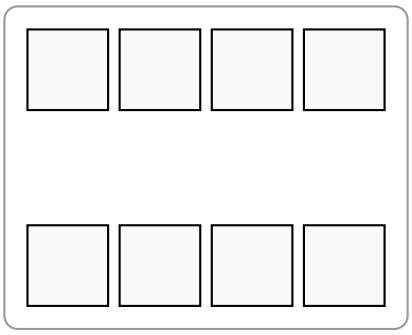
A



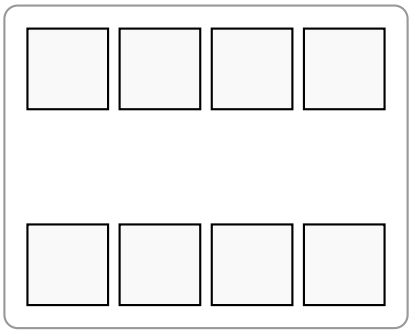
B



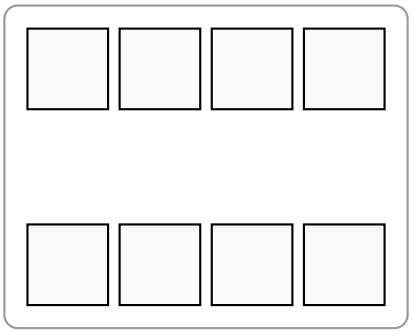
A then B



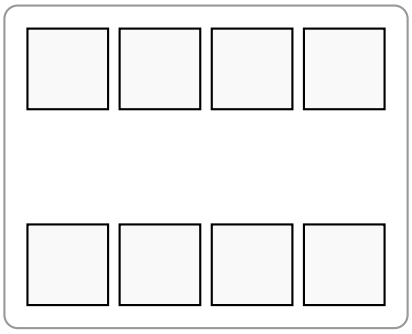
undoes A



undoes B

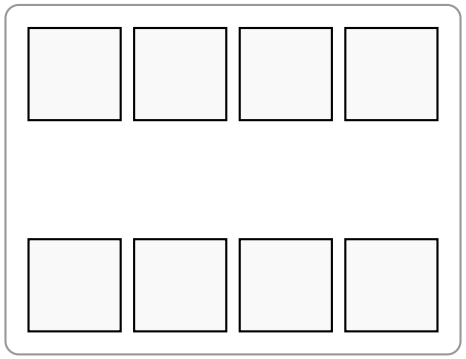


undoes (A then B)

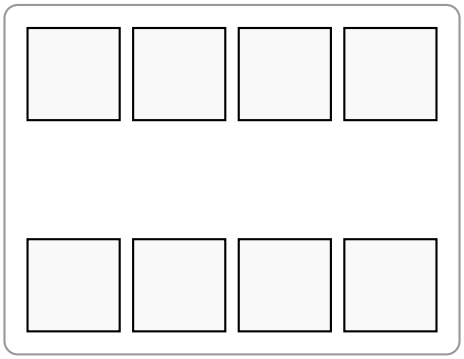


Problem 8: Find two different machines A and B with 4 slots, each moving all four cubes, so that A then B is the same machine as B then A.

A



B



Problem 9: For each number of slots from 1 to 10, find the largest number of turns a machine can take to bring all the cubes home. Can you always build a slower machine when you have one more slot? Explain.

slots	1	2	3	4	5	6	7	8	9	10
most turns										

Problem 10: How many perfect shuffles bring a full deck of 52 cards back to its starting order? Explain how you know.